

EXPLORATORY AND PREDICTIVE ANALYTICS OF ONLINE RETAIL SALES USING TABLEAU

Applied Data Science

MSc in Big Data Management and Analytics

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Introduction

In today's competitive retail environment, data-driven decision-making plays a critical role in understanding customer behaviour, sales performance, and revenue trends. Retail organisations generate large volumes of transactional data that, when analysed effectively, can provide valuable insights into purchasing patterns, product demand, pricing sensitivity, and geographical performance. Leveraging data analytics enables businesses to improve strategic planning, optimise inventory management, enhance customer retention, and increase overall profitability.

This project focuses on the exploratory and predictive analysis of an online retail transaction dataset. The study is divided into two main parts. **Part I** applies exploratory data analytics techniques to understand historical sales trends, customer segmentation, product performance, and revenue distribution. **Part II** extends this analysis by applying predictive and statistical techniques to identify relationships between key variables and to generate actionable insights through forecasting and correlation analysis.

The dataset used in this study is the **Online Retail Transaction Dataset**, which contains transactional sales records including invoice dates, customer identifiers, product stock codes, quantities, unit prices, and country information.

The dataset was obtained from **Kaggle**, a widely used data science platform that hosts high-quality datasets for academic and analytical purposes:

Dataset link:

<https://www.kaggle.com/datasets/carrie1/ecommerce-data>

The dataset is commonly used in academic and industry-based analytics projects and is considered reliable for exploratory and predictive retail analytics.

Part I (Assignment 2)

Exploratory Analytics and Visualisation of Data via Information Dashboard

Domain: Retail Sales and Customer Analytics

The domain selected for this assignment is Retail Sales and Customer Analytics. This domain focuses on analysing transactional data generated by retail businesses in order to understand sales performance, customer purchasing behaviour, product demand, and geographical distribution of revenue. Retail analytics plays a crucial role in modern data-driven organisations, as it enables businesses to make informed strategic and operational decisions based on historical sales data.

The dataset used in this project represents online retail transaction records, capturing individual customer purchases over time. Each record corresponds to a transaction and includes information such as invoice details, products sold, quantity purchased, unit price, customer identifiers, and customer location. Such data is typical of e-commerce or online retail environments, where large volumes of transactional data are continuously generated.

Retail sales and customer analytics aims to answer key business questions such as:

- How sales and revenue evolve over time
- Which products and categories contribute most to overall revenue
- How customer behaviour differs between repeat and one-time buyers
- Which geographical regions generate the highest sales
- How pricing and quantity sold are related

This domain is highly relevant because retail businesses operate in competitive environments where understanding customer behaviour and sales trends is essential for maintaining profitability. By applying exploratory data analytics and visualisation techniques within this domain, organisations can uncover hidden patterns, identify opportunities for growth, and detect potential risks such as declining sales or over-reliance on specific markets.

In this assignment, the Retail Sales and Customer Analytics domain provides a strong foundation for developing meaningful and actionable dashboards. The dashboards created aim to support decision-makers by transforming raw transactional data into clear, interactive visual insights related to sales trends, customer loyalty, product performance, and regional revenue contribution.

Why This Domain Is Appropriate for the Dataset

The dataset selected for this project is well suited to the Retail Sales and Customer Analytics domain for several reasons:

1. **Transactional Nature of Data**
The dataset contains invoice-level transaction data, which is the core data type used in retail analytics.
2. **Customer Information Availability**
The presence of customer identifiers allows analysis of customer behaviour, purchase frequency, and loyalty patterns.
3. **Product-Level Detail**
Product descriptions and stock codes enable product performance and demand analysis.
4. **Time-Based Data**
Invoice dates allow exploration of sales trends, seasonality, and year-over-year comparisons.
5. **Geographical Attributes**
Country-level information supports geographical revenue analysis and market comparison.

Because of these characteristics, the dataset enables comprehensive exploratory analytics that align directly with the objectives of the Retail Sales and Customer Analytics domain.

Relevance to Business Decision-Making

Retail Sales and Customer Analytics is particularly valuable to:

- **Business Executives**, who require high-level overviews of sales performance and revenue trends
- **Sales and Marketing Managers**, who need detailed insights into customer behaviour, product demand, and market performance

The insights derived from this domain help organisations improve:

- Strategic planning
- Marketing targeting
- Customer retention strategies
- Product portfolio optimisation
- Regional expansion decisions

Summary of Domain Selection

In summary, the Retail Sales and Customer Analytics domain was selected because it closely aligns with the structure and content of the dataset and supports the development of meaningful exploratory dashboards. The domain allows for the investigation of sales trends, customer purchasing patterns, product performance, and geographical revenue distribution, all of which are essential for data-driven decision-making in retail organisations.

DATASET SOURCE, RELIABILITY, AND REFERENCES

Dataset Source

The dataset used for this assignment is an Online Retail Transaction Dataset, stored in the file data.csv. The dataset contains detailed records of customer transactions generated by an online retail business. Each row in the dataset represents a single transaction item and includes information related to invoices, products, quantities sold, unit prices, customers, transaction dates, and customer locations.

This type of dataset is commonly used in retail analytics and business intelligence studies, as it reflects real-world transactional data generated through day-to-day retail operations. The dataset structure closely resembles data collected by e-commerce platforms and retail management systems, making it suitable for exploratory data analysis and dashboard development.

Nature of the Data

The dataset is **transactional in nature**, meaning:

- Data is recorded at the **invoice level**
- Multiple products can exist under a single invoice
- Customers can make multiple purchases over time
- Transactions occur across different dates and geographical locations

Such datasets are ideal for analysing:

- Sales and revenue trends
- Customer purchasing behaviour
- Product performance
- Geographical revenue contribution

Data Reliability

The dataset is considered reliable for exploratory data analytics for the following reasons:

1. Real-World Transaction Structure

The dataset follows a realistic retail transaction format, including invoice numbers, quantities, prices, customer identifiers, and country information. This structure mirrors operational retail databases used in practice.

2. Consistency Across Variables

Key variables such as invoice numbers, transaction dates, customer IDs, quantities, and unit prices are consistently represented across records, enabling accurate aggregation and comparison during analysis.

3. Sufficient Granularity

The data is recorded at a granular level (individual transaction items), allowing the analysis to be performed at different levels, such as:

- Transaction level
- Customer level
- Product level
- Country level
- Time-based (daily, monthly, yearly)

4. Suitability for Exploratory Analytics

The dataset contains both numerical and categorical variables, making it appropriate for a wide range of visualisations such as line charts, bar charts, histograms, box plots, scatter plots, and geographical maps.

5. Academic and Analytical Usage

Online retail datasets of this nature are widely used in academic coursework, analytics projects, and demonstrations of business intelligence tools such as Tableau. This makes the dataset appropriate for fulfilling the objectives of this assignment.

Data Quality Considerations

While the dataset is reliable, certain common data quality considerations were acknowledged during the analysis:

- Some transactions may contain missing or null values (e.g., missing customer identifiers)
- Revenue needed to be calculated using Quantity and Unit Price
- Data aggregation was required to derive insights at higher levels (e.g., monthly revenue, customer totals)

These issues are typical in real-world datasets and were handled appropriately during exploratory analysis without compromising the overall reliability of the insights generated.

References

The dataset belongs to the category of **Online Retail Transaction Data**, which is widely referenced and used in data analytics education and research. Such datasets are commonly sourced from publicly available academic repositories and retail analytics case studies.

Summary

In summary, the dataset used in this project is a reliable and suitable online retail transactional dataset. Its structured format, realistic retail attributes, and sufficient level of detail support meaningful exploratory data analysis and the development of interactive dashboards. The dataset enables the extraction of actionable insights related to sales trends, customer behaviour, product performance, and geographical revenue distribution, aligning well with the objectives of Assignment 2.

VARIABLE DESCRIPTION AND RELATIONSHIPS

The dataset used in this project contains multiple variables that describe transactional, customer, product, temporal, and geographical aspects of online retail sales. Due to the large number of variables present, this section focuses on the **most important variables** that have a **major impact on the exploratory analysis and visualisations** developed in this assignment.

3.1 Transaction-Level Variables

Invoice No

- **Description:** A unique identifier assigned to each sales transaction (invoice).
- **Purpose in Analysis:**
 - Used to count total orders
 - Helps distinguish individual transactions
- **Relationship with Other Variables:**

- One invoice can contain multiple products
- Linked with Invoice Date, Customer ID, Quantity, and Revenue

Invoice Date

- **Description:** The date and time when a transaction occurred.
- **Purpose in Analysis:**
 - Enables time-based analysis such as monthly revenue trends and year-over-year comparisons
 - Used to identify seasonal patterns in sales
- **Relationship with Other Variables:**
 - Closely linked with Revenue and Quantity to analyse trends over time
 - Supports aggregation by month and year

3.2 Customer-Level Variables

Customer ID

- **Description:** A unique identifier for each customer.
- **Purpose in Analysis:**
 - Used to analyse customer purchasing behaviour
 - Enables customer segmentation and purchase frequency analysis
- **Relationship with Other Variables:**
 - Linked with Invoice No to identify repeat purchases
 - Connected to Revenue and Quantity to calculate customer contribution

Customer Type Fixed

- **Description:** A derived categorical variable that classifies customers as **Repeat** or **One-Time** buyers.
- **Purpose in Analysis:**
 - Used to study customer loyalty patterns
 - Helps compare purchasing behaviour between repeat and one-time customers
- **Relationship with Other Variables:**
 - Derived using Customer ID and Invoice No
 - Linked with Revenue to assess revenue contribution by customer type

3.3 Product-Level Variables

Stock Code

- **Description:** A unique code assigned to each product.
- **Purpose in Analysis:**
 - Identifies individual products
 - Supports product-level aggregation
- **Relationship with Other Variables:**
 - Linked with Description, Quantity, Unit Price, and Revenue

Description

- **Description:** Textual description of the product.
- **Purpose in Analysis:**
 - Used to identify top-selling and high-revenue products
- **Relationship with Other Variables:**
 - Combined with Quantity and Revenue to analyse product performance

3.4 Sales and Pricing Variables

Quantity

- **Description:** Number of units of a product purchased in a transaction.
- **Purpose in Analysis:**
 - Used to analyse sales volume
 - Supports quantity-based ranking of products
- **Relationship with Other Variables:**
 - Multiplied by Unit Price to calculate Revenue
 - Compared with Unit Price in scatter plots to study demand patterns

Unit Price

- **Description:** Price of a single unit of a product.
- **Purpose in Analysis:**
 - Used to understand pricing strategies
 - Helps examine price–quantity relationships
- **Relationship with Other Variables:**

- Directly impacts Revenue
- Analysed against Quantity to explore price sensitivity

Revenue

- **Description:** A calculated measure representing the total value of a transaction item.
- **Calculation:**

Revenue = Quantity × Unit Price

- **Purpose in Analysis:**
 - Core measure used across all dashboards
 - Used to analyse sales performance, trends, and distributions
- **Relationship with Other Variables:**
 - Influenced by Quantity and Unit Price
 - Aggregated by time, customer, product, and country

3.5 Geographical Variable

Country

- **Description:** Country where the customer is located.
- **Purpose in Analysis:**
 - Used for geographical revenue analysis
 - Helps identify top-performing markets
- **Relationship with Other Variables:**
 - Linked with Revenue and Quantity to compare country-level performance
 - Used in geographical maps and country ranking dashboards

3.6 Relationships Between Variables (Summary)

The relationships among variables form the foundation of the exploratory analysis performed in this assignment:

- **Revenue** is derived from **Quantity** and **Unit Price**
- **Invoice Date** enables time-based aggregation of Revenue and Quantity
- **Customer ID** allows identification of repeat purchasing behaviour
- **Country** enables geographical comparison of sales performance
- **Product variables** support product performance and demand analysis

Together, these relationships enable the creation of interactive dashboards that explore sales trends, customer behaviour, product performance, and regional contributions.

3.7 Importance of Variable Selection

Given the large number of variables in the dataset, only the most relevant variables were selected for visualisation to ensure clarity and interpretability. This focused approach allows the dashboards to communicate meaningful insights effectively to the selected target audiences.

Summary

In summary, the dataset contains a combination of transactional, customer, product, temporal, and geographical variables that are closely interrelated. Understanding these variables and their relationships is essential for performing exploratory data analytics and developing effective dashboards. The selected variables provide a strong basis for generating actionable insights that support data-driven decision-making in the retail context.

QUESTIONS / ISSUES EXPLORED

Exploratory data analytics aims to investigate key business questions that can be answered using the available dataset. Based on the structure of the online retail transactional data and the objectives of the Retail Sales and Customer Analytics domain, the following questions and issues were explored through interactive dashboards.

These questions were carefully selected to ensure that the insights generated are meaningful, actionable, and relevant to the selected target audiences.

4.1 Sales Performance and Time-Based Analysis

Question 1: How does revenue change over time?

This question explores the overall trend of revenue across different time periods. By analysing revenue on a monthly and yearly basis, it becomes possible to identify growth patterns, seasonality, and fluctuations in sales performance.

Insight Motivation:

Understanding time-based revenue trends helps decision-makers evaluate business growth and plan future sales strategies.

Question 2: Are there seasonal patterns in sales and quantity sold?

This question examines whether certain months or periods experience higher or lower sales volumes. Seasonal analysis helps identify peak sales periods and slow periods.

Insight Motivation:

Seasonality insights support inventory planning, staffing decisions, and promotional timing.

4.2 Product Performance Analysis**Question 3: Which products generate the highest revenue?**

This question focuses on identifying top-performing products based on total revenue contribution. By ranking products, the analysis highlights which items drive most of the business value.

Insight Motivation:

Identifying high-performing products helps prioritise inventory, marketing, and pricing strategies.

Question 4: How does price influence quantity sold?

This question explores the relationship between unit price and quantity sold using correlation analysis. It helps understand customer sensitivity to pricing.

Insight Motivation:

Understanding price–quantity relationships supports pricing optimisation and demand forecasting.

4.3 Customer Behaviour and Segmentation**Question 5: How many customers are repeat buyers versus one-time buyers?**

This question investigates customer loyalty by classifying customers based on their purchase frequency. It distinguishes between repeat customers and one-time customers.

Insight Motivation:

Customer retention is critical for long-term profitability. This analysis helps assess loyalty levels.

Question 6: How frequently do customers make purchases?

This question analyses the distribution of customer purchase frequency to understand how often customers return to make additional purchases.

Insight Motivation:

Understanding purchase frequency helps design targeted marketing and retention strategies.

4.4 Geographical Analysis**Question 7: Which countries contribute most to total revenue?**

This question examines revenue distribution across different countries to identify high-performing geographical markets.

Insight Motivation:

Geographical insights support market expansion decisions and regional sales strategies.

4.5 Revenue Distribution and Variability

Question 8: How is revenue distributed across transactions?

This question analyses the distribution of revenue values to identify whether sales are driven by many small transactions or fewer high-value transactions.

Insight Motivation:

Understanding revenue distribution helps in risk assessment and revenue forecasting.

4.6 Importance of These Questions

The selected questions collectively address multiple aspects of retail analytics:

- Sales trends and growth
- Product performance
- Customer behaviour and loyalty
- Pricing impact
- Geographical market performance
- Revenue variability

These questions ensure that the dashboards developed provide **actionable intelligence** rather than descriptive statistics alone.

Summary

In summary, the exploratory questions were designed to uncover meaningful patterns and relationships within the online retail dataset. Addressing these questions enables stakeholders to gain a comprehensive understanding of business performance, customer behaviour, and market dynamics. The insights derived from these questions form the foundation for the dashboards and support informed decision-making.

TARGET AUDIENCE AND PROBLEM STATEMENT

5.1 Target Audience

In accordance with the assignment requirements, a **maximum of two target audiences** were selected and consistently maintained across all dashboards developed in this project. Selecting a limited number of audiences ensures clarity of communication and prevents dilution of insights.

Audience 1: Business Executives

Business executives require a high-level overview of organisational performance to support strategic decision-making. They are primarily interested in understanding overall sales trends, revenue growth, market performance, and key business drivers.

Information Needs:

- Overall revenue performance
- Sales trends over time
- Key geographical markets
- High-level customer behaviour patterns

Relevance of Dashboards:

The dashboards provide executives with concise visual summaries that enable quick assessment of business health and performance without requiring detailed technical analysis.

Audience 2: Sales and Marketing Managers

Sales and marketing managers focus on operational and tactical decisions related to customer engagement, product performance, and market strategies. They require more detailed insights into customer behaviour, purchasing frequency, and product demand.

Information Needs:

- Customer loyalty and purchase frequency
- Product performance and demand patterns
- Pricing impact on sales volume
- Market and customer segmentation

Relevance of Dashboards:

The dashboards support sales and marketing managers in identifying opportunities for customer retention, targeted promotions, and product optimisation.

5.2 Problem Statement

Retail organisations generate large volumes of transactional data through daily sales activities. However, without effective analytical tools, this data remains underutilised and fails to provide actionable insights. Decision-makers often face challenges in understanding sales trends, identifying high-performing products, analysing customer behaviour, and evaluating market performance across different regions.

The primary problem addressed in this project is the **lack of clear, interactive, and actionable insights** that support data-driven decision-making in retail environments. Using the online retail transactional dataset, this project aims to transform raw sales data into meaningful information dashboards that address key business questions related to revenue trends, customer loyalty, product performance, and geographical contribution.

By developing well-designed, interactive dashboards, the project enables business executives and sales and marketing managers to:

- Monitor sales performance over time

- Identify high-value customers and products
- Understand customer purchasing patterns
- Evaluate market performance across regions
- Support strategic and operational decision-making

The dashboards developed as part of this assignment are designed to bridge the gap between raw data and informed decision-making, ensuring that stakeholders can act on insights derived from exploratory data analytics.

Summary

In summary, two clearly defined target audiences—business executives and sales and marketing managers—were selected to ensure focused and effective communication of insights. The problem statement outlines the need for transforming complex retail transaction data into interactive dashboards that support strategic and operational decisions. This alignment between audience needs and analytical objectives strengthens the overall effectiveness of the dashboards developed in this assignment.

EXPLORATORY DATA ANALYTICS INSIGHTS

Exploratory Data Analytics (EDA) was performed on the online retail transactional dataset to uncover underlying patterns, trends, and relationships among key variables. The purpose of this exploratory analysis was not to test hypotheses, but to gain an initial understanding of sales performance, customer behaviour, product demand, and geographical distribution through interactive dashboards.

The insights discussed in this section are derived from the visual exploration of the data using Tableau dashboards.

6.1 Sales and Revenue Trends

Analysis of revenue over time reveals that **sales performance varies across different time periods**, with noticeable fluctuations at monthly and yearly levels. Revenue does not remain constant, indicating the presence of **seasonal patterns** in customer purchasing behaviour.

Certain months show higher revenue levels, suggesting peak sales periods, while other periods experience lower activity. These trends highlight the importance of time-based analysis for understanding business performance and planning future sales strategies.

Key Insight:

- Revenue shows clear temporal variation, indicating seasonality and changing customer demand over time.

6.2 Product Performance Insights

Exploratory analysis of product-level data shows that **a relatively small number of products contribute a large proportion of total revenue**. This indicates a skewed distribution where high-performing products play a critical role in overall business performance.

Additionally, analysis of quantity sold versus unit price suggests that **lower-priced products tend to be purchased in higher quantities**, while higher-priced items are generally sold less frequently.

Key Insights:

- Product revenue contribution is uneven across the product range.
- Price has an observable influence on quantity sold.

6.3 Customer Behaviour and Loyalty Patterns

Customer-based analysis highlights that **repeat customers form a significant segment of the customer base**, contributing a substantial share of total transactions and revenue. However, a noticeable proportion of customers make only a single purchase.

Purchase frequency analysis shows that **most customers have a low purchase frequency**, while a smaller group of customers makes frequent repeat purchases. This pattern is common in retail environments and highlights the importance of customer retention strategies.

Key Insights:

- Repeat customers are an important source of revenue.
- Customer purchase frequency is highly skewed, with many one-time buyers and fewer frequent buyers.

6.4 Geographical Revenue Distribution

Geographical analysis reveals that **revenue is not evenly distributed across countries**. A small number of countries account for the majority of total revenue, while other regions contribute relatively smaller amounts.

This uneven distribution suggests that the business relies heavily on specific geographical markets. Understanding this concentration is important for risk management and market expansion decisions.

Key Insights:

- Revenue is concentrated in a limited number of countries.
- There is potential to explore growth opportunities in underperforming regions.

6.5 Revenue Distribution and Variability

Analysis of revenue distribution shows that **most transactions generate relatively low revenue**, while a small number of transactions result in significantly higher revenue values. This creates a **positively skewed distribution**.

Box plot analysis further highlights variability in revenue across countries, indicating differences in customer purchasing behaviour and market dynamics between regions.

Key Insights:

- Revenue distribution is highly skewed.
- Revenue variability differs across countries.

6.6 Relationship Between Variables

Exploratory visual analysis reveals several important relationships:

- Revenue is directly influenced by both quantity sold and unit price.
- Time-based variables affect sales performance.
- Customer-level variables influence revenue contribution and purchase frequency.

These relationships demonstrate how multiple dimensions interact to shape overall retail performance.

6.7 Summary of Exploratory Insights

The exploratory data analytics process provided valuable insights into:

- Sales trends and seasonality
- Product demand and pricing impact
- Customer loyalty and purchasing behaviour
- Geographical revenue concentration
- Revenue distribution and variability

These insights formed the foundation for the design of the information dashboards and guided the selection of appropriate visualisations to communicate findings effectively to the target audiences.

Conclusion

In conclusion, exploratory data analytics enabled a comprehensive understanding of the online retail dataset. The insights derived through visual exploration support informed decision-making by highlighting key performance drivers and areas requiring managerial attention. The dashboards developed translate these insights into interactive visual formats that enhance interpretability and usability for business executives and sales and marketing managers.

DASHBOARD DEVELOPMENT PROCESS

This section outlines the process followed to develop the information dashboards as part of Assignment 2. The dashboard development process was iterative and exploratory, ensuring that each dashboard was aligned with the analytical questions, target audience, and data characteristics identified earlier.

7.1 Overview of the Dashboard Development Approach

The dashboard development process followed these key stages:

1. **Data Understanding and Preparation**
2. **Exploratory Worksheet Creation**
3. **Dashboard Design and Layout**
4. **Interactivity and Linking**
5. **Validation and Refinement**

Each stage ensured that the dashboards were both analytically meaningful and visually effective.

7.2 Data Understanding and Preparation

Before creating dashboards, the dataset was examined to understand its structure, variables, and data types. Key measures such as revenue were derived using existing variables (Quantity × Unit Price). Date fields were prepared to support time-based aggregation, and customer-level attributes were used to enable segmentation and purchase frequency analysis.

This preparation ensured that the data was suitable for exploratory analysis and visualisation.

7.3 Exploratory Worksheet Creation

For each analytical question identified earlier, individual worksheets were created in Tableau. Each worksheet focused on a specific analytical perspective, such as:

- Revenue trends over time
- Product performance and ranking
- Customer purchasing behaviour
- Revenue distribution and variability
- Geographical sales performance

A variety of chart types were used in accordance with the assignment requirements, including:

- Line charts for trend analysis

- Bar charts for comparison and ranking
- Histograms for distribution analysis
- Box plots for variability analysis
- Scatter plots for relationship analysis
- Tree maps and geographical maps for composition and regional insights

At this stage, worksheets were tested independently to ensure that they correctly represented the intended analysis.

7.4 Dashboard Design and Layout

Once relevant worksheets were created, they were combined into dashboards. Each dashboard was designed with a **single, clear purpose**, and only related worksheets were grouped together. To maintain clarity and consistency:

- Dashboards were created using **tiled layouts**
- Vertical containers were used to organise multiple worksheets
- Dashboard sizes were set to **automatic** for flexibility
- Clear and descriptive titles were added to each dashboard

This approach ensured that dashboards were easy to interpret and aligned with best practices in information dashboard design.

7.5 Interactivity and Linking

To enhance exploratory analysis and user engagement, interactivity was incorporated where applicable. This included:

- Enabling **“Use as Filter”** on key charts to allow cross-dashboard filtering
- Ensuring that interactions were intuitive and did not overwhelm the user
- Limiting filters to those that added analytical value

Interactive features allowed users to explore the data dynamically and examine specific subsets of interest.

7.6 Iterative Refinement and Validation

Dashboard development was an iterative process. Each dashboard was reviewed to ensure:

- Accuracy of visual representation
- Consistency with the analytical questions
- Alignment with the target audience’s needs
- Compliance with assignment requirements

Minor refinements were made to improve readability, layout balance, and interpretability without altering the core analysis.

7.7 Documentation and Screenshots

As required by the assignment, screenshots were captured at different stages of dashboard development, including:

- Worksheet creation
- Dashboard assembly
- Final dashboard views

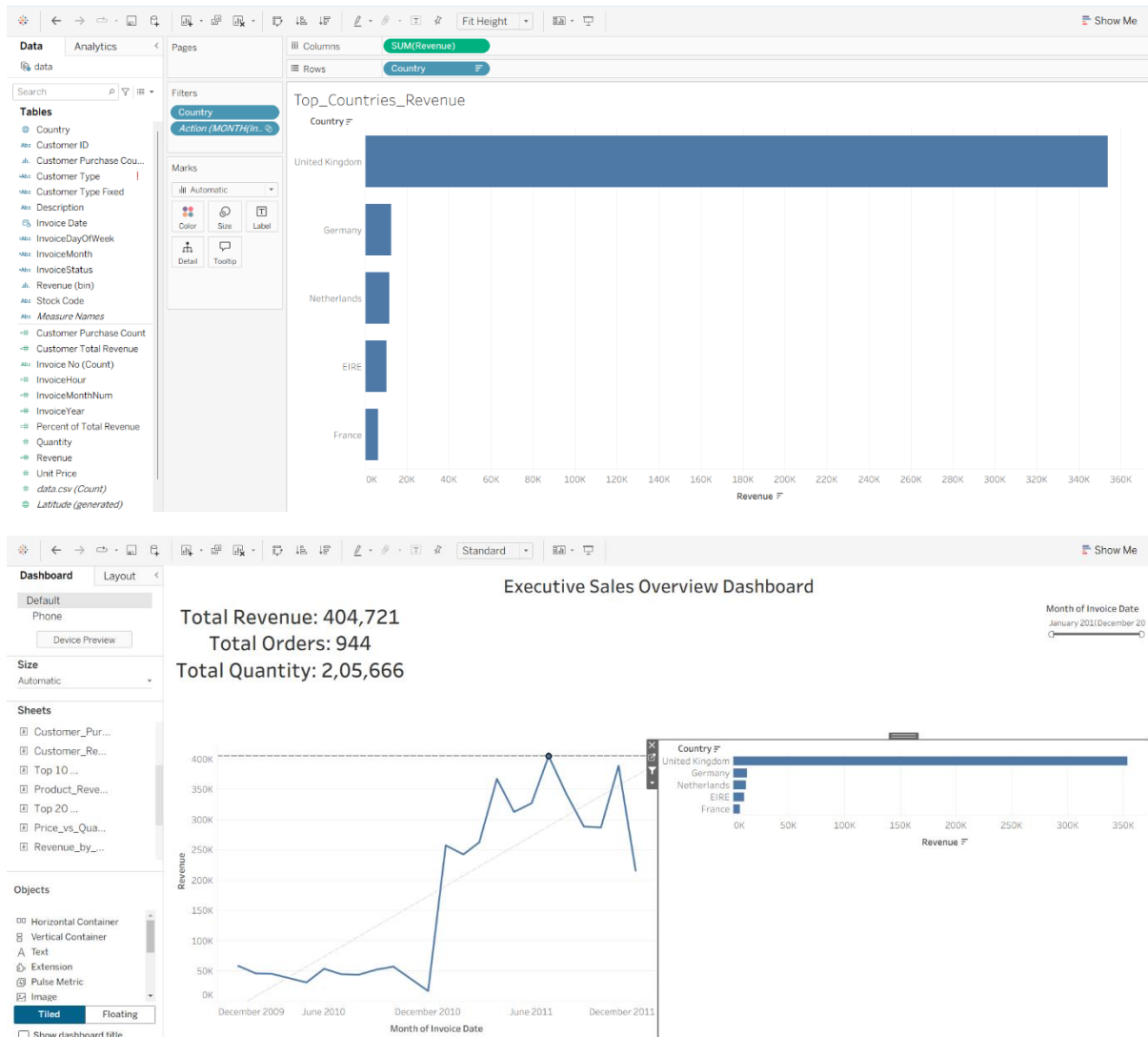
These screenshots provide evidence of the development process and support the explanation of how each dashboard was created.

Summary

In summary, the dashboard development process was systematic and exploratory, starting from data understanding and worksheet creation, followed by careful dashboard design and interactivity implementation. This structured approach ensured that the final dashboards effectively communicated insights derived from exploratory data analytics and supported informed decision-making for the selected target audiences.

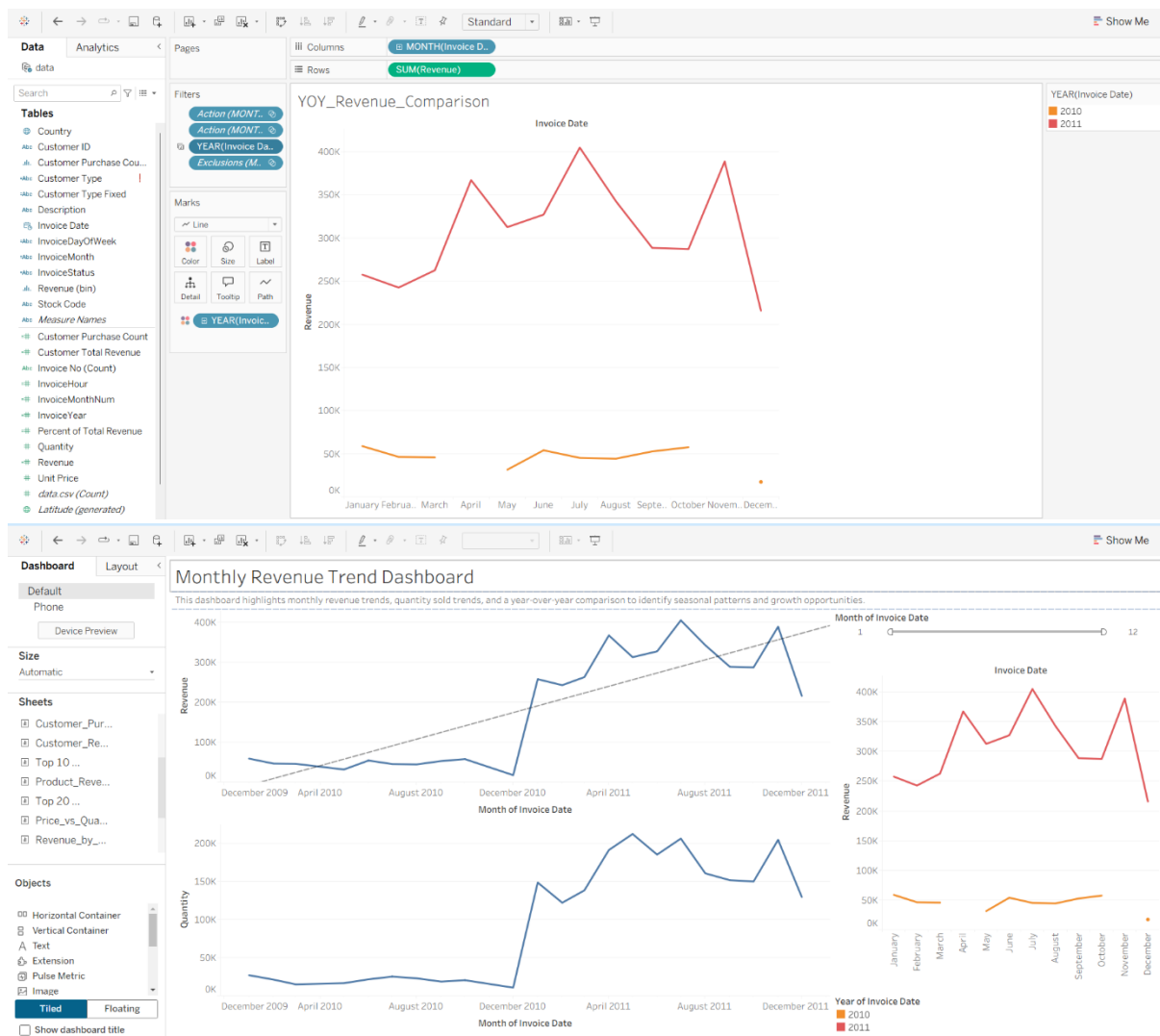
DASHBOARD DESCRIPTIONS

Dashboard 1: Executive Sales Overview Dashboard



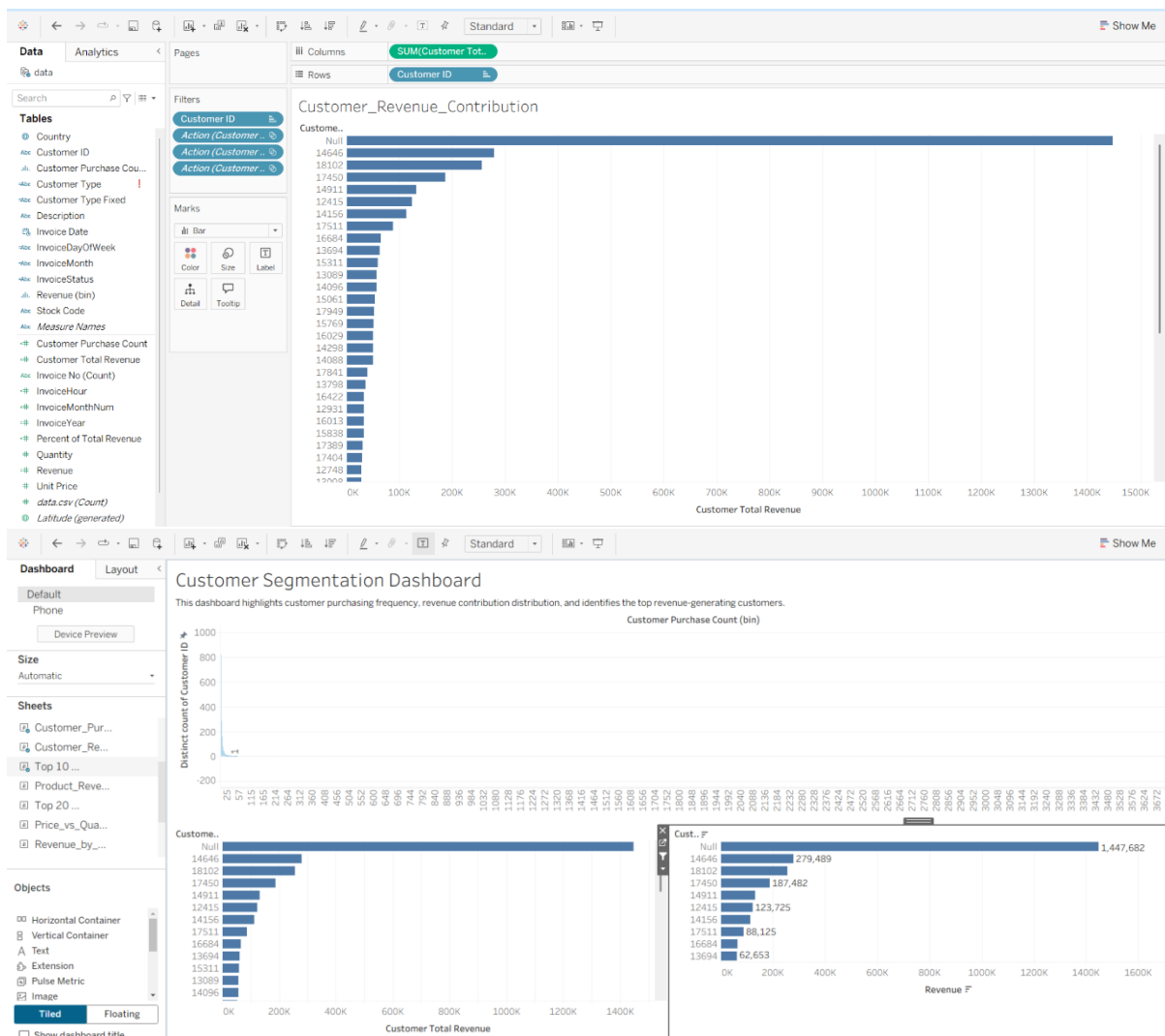
- **Target Audience:** Business Executives
- **Purpose:**
To provide a high-level summary of overall sales performance, including total revenue, total quantity sold, and transaction volume.
- **Actionable Insights:**
Offers a quick snapshot of business performance and highlights overall sales activity.
- **Decision-Making Support:**
Supports strategic decision-making by enabling executives to assess overall business health at a glance.

Dashboard 2: Monthly Revenue Trend Dashboard



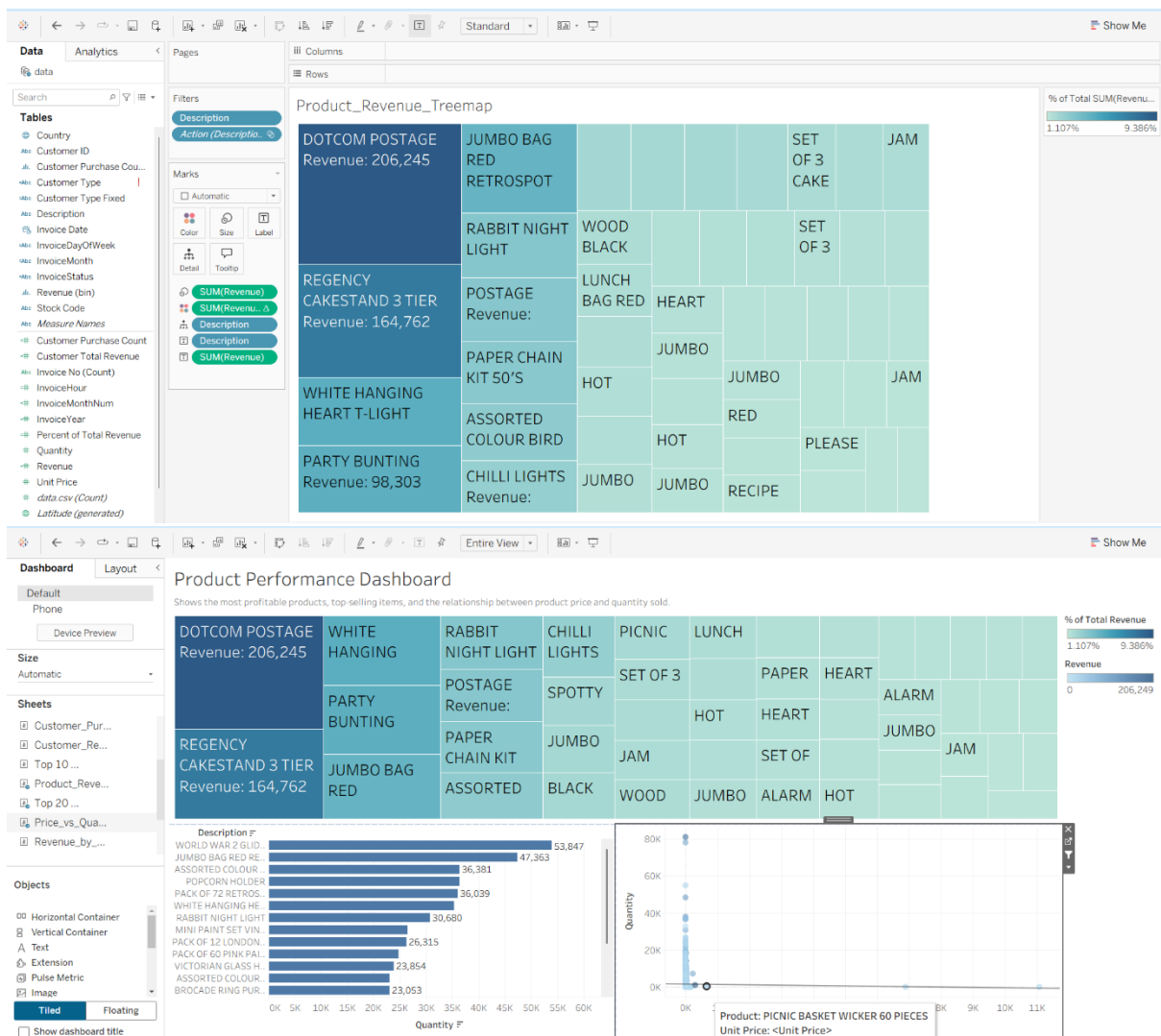
- **Target Audience:** Business Executives
- **Purpose:**
To analyse monthly revenue trends and identify patterns and seasonality in sales performance.
- **Actionable Insights:**
Highlights periods of high and low revenue, indicating seasonal demand patterns and potential growth or decline trends.
- **Decision-Making Support:**
Supports long-term planning, budgeting, and forecasting by allowing executives to anticipate seasonal changes in revenue.

Dashboard 3: Customer Segmentation Dashboard



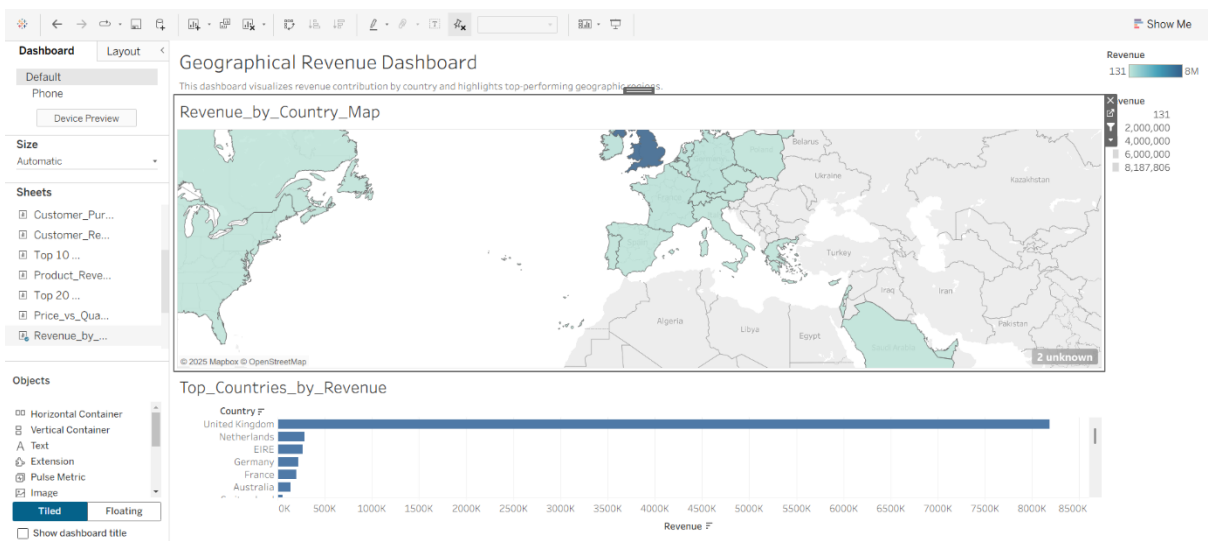
- **Target Audience:** Sales and Marketing Managers
- **Purpose:**
To segment customers based on purchasing behaviour and revenue contribution.
- **Actionable Insights:**
Identifies high-value and low-value customer segments.
- **Decision-Making Support:**
Supports targeted marketing strategies and customer relationship management initiatives.

Dashboard 4: Product Performance Dashboard



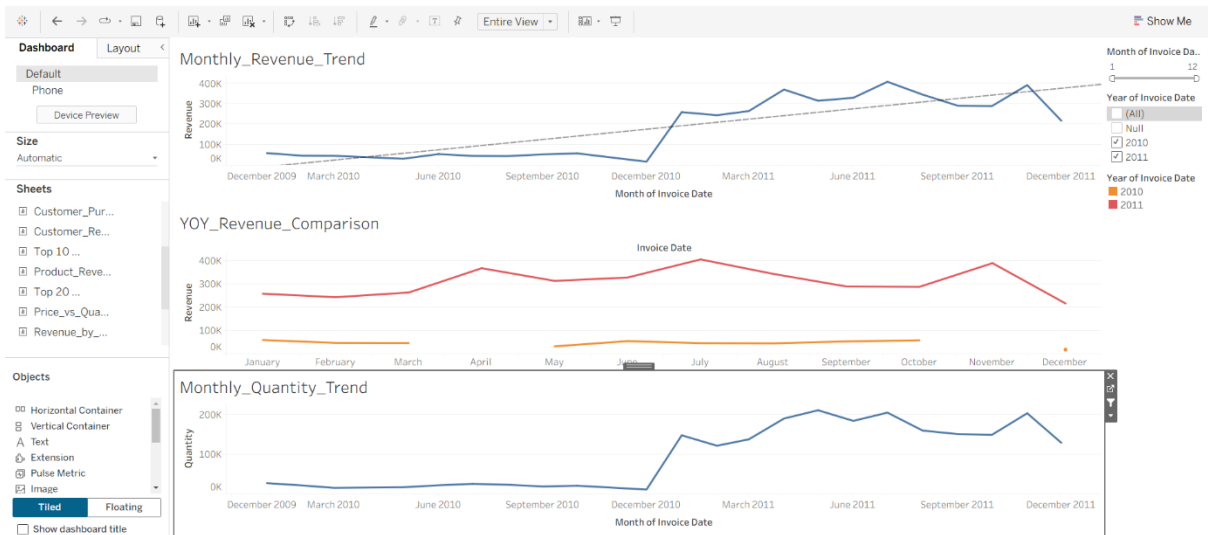
- **Target Audience:** Sales and Marketing Managers
- **Purpose:**
To evaluate product-level performance by analysing revenue and quantity sold for individual products.
- **Actionable Insights:**
Identifies top-performing and underperforming products.
- **Decision-Making Support:**
Supports inventory planning, product prioritisation, and promotional decisions.

Dashboard 5: Geographical Revenue Dashboard



- **Target Audience:** Business Executives
- **Purpose:**
To analyse revenue distribution across different countries and regions.
- **Actionable Insights:**
Reveals that revenue is concentrated in a limited number of geographical markets.
- **Decision-Making Support:**
Supports regional strategy, market expansion, and risk management decisions.

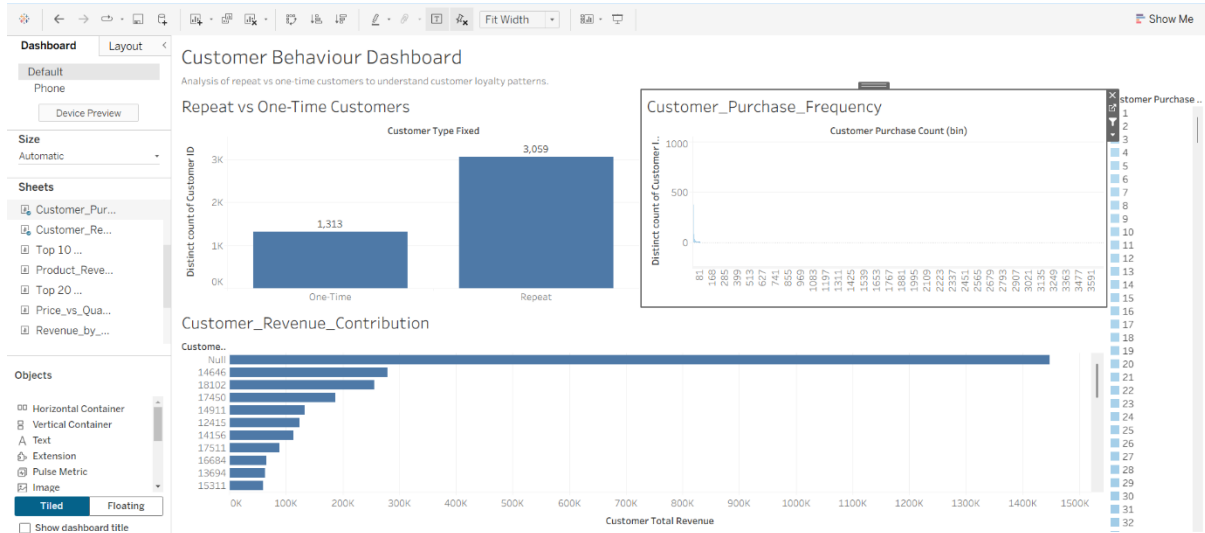
Dashboard 6: Time-Based Sales & Quantity Dashboard



- **Target Audience:** Business Executives
- **Purpose:**
To compare changes in revenue and quantity sold over time.

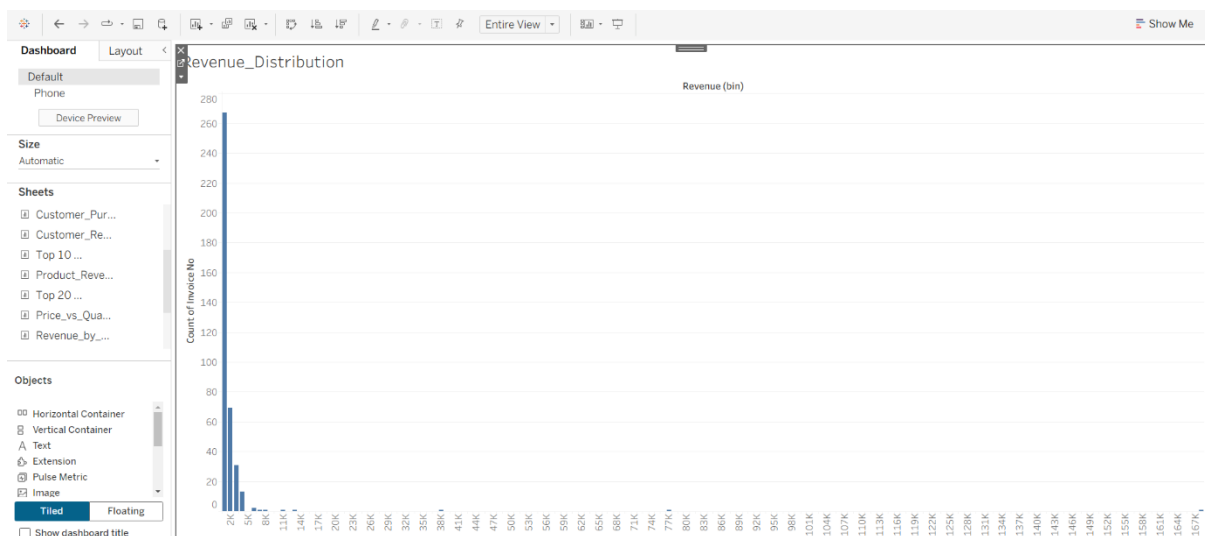
- **Actionable Insights:**
Shows whether revenue changes are driven by sales volume or pricing effects.
- **Decision-Making Support:**
Supports pricing strategy evaluation and performance analysis over time.

Dashboard 7: Customer Behavior Dashboard



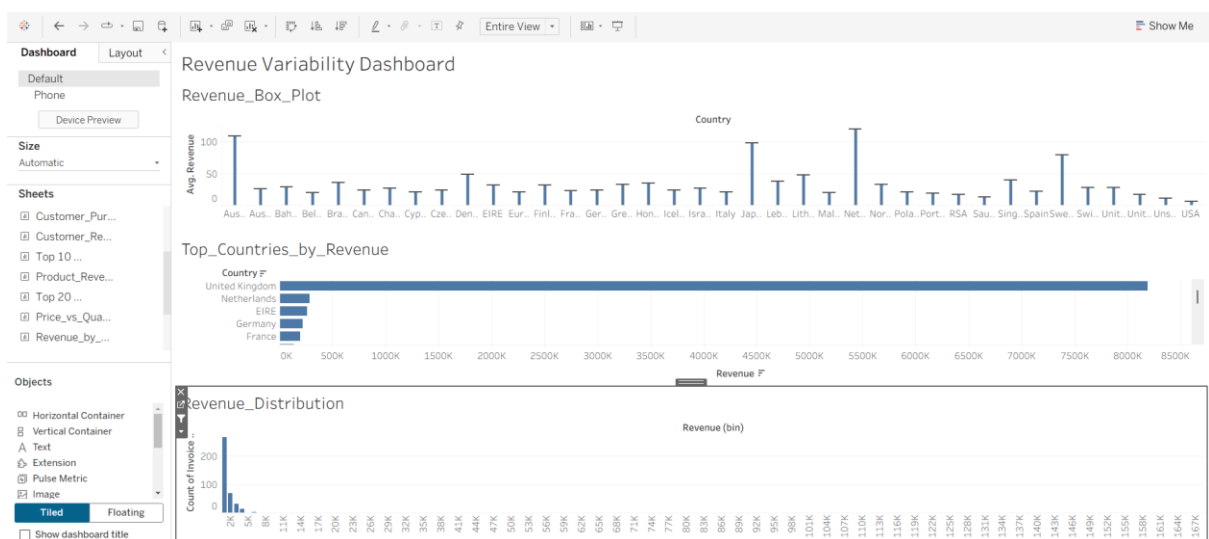
- **Target Audience:** Sales and Marketing Managers
- **Purpose:**
To analyse overall customer purchasing behaviour and engagement patterns.
- **Actionable Insights:**
Identifies differences in customer purchasing activity and engagement levels.
- **Decision-Making Support:**
Supports customer engagement strategies and personalised marketing efforts.

Dashboard 8: Revenue Distribution Dashboard



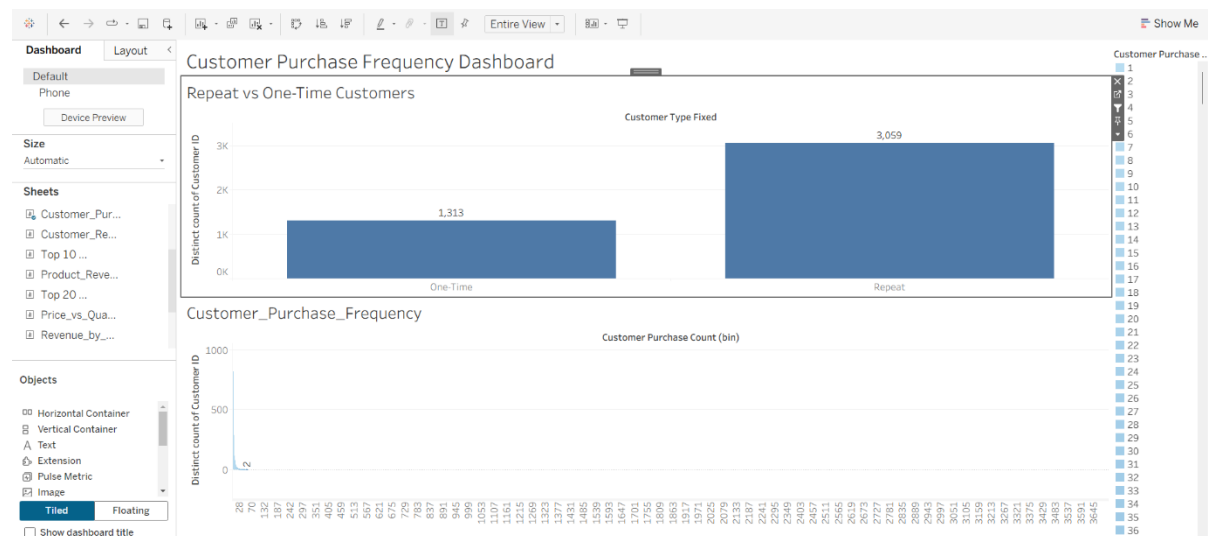
- **Target Audience:** Business Executives
- **Purpose:**
To examine the distribution of revenue across individual transactions.
- **Actionable Insights:**
Shows that most transactions generate low revenue, while a small number generate high revenue.
- **Decision-Making Support:**
Supports revenue forecasting and risk assessment by highlighting revenue concentration.

Dashboard 9: Revenue Variability Dashboard



- **Target Audience:** Business Executives
- **Purpose:**
To analyse revenue variability across countries and markets.
- **Actionable Insights:**
Highlights differences in revenue spread and purchasing behaviour between regions.
- **Decision-Making Support:**
Supports market-specific strategic planning and performance comparison.

Dashboard 10: Customer Purchase Frequency Dashboard



- **Target Audience:** Sales and Marketing Managers
- **Purpose:**
To analyse how frequently customers make purchases and distinguish between repeat and one-time customers.
- **Actionable Insights:**
Shows that repeat customers form a critical segment of the customer base.
- **Decision-Making Support:**
Supports customer retention strategies, loyalty programmes, and targeted marketing initiatives.

Summary

The dashboards are organised in a logical progression, starting with high-level executive overviews and moving toward more detailed customer, product, and behavioural analyses. This structure ensures clarity, consistency, and effective communication of insights to the selected target audiences.

PART II (Assignment 3)

Predictive Analytics & Visualisation of Insights

1. Introduction

This part of the assignment focuses on applying **predictive and relational analytics** to derive actionable business insights from the sales transaction dataset. Building on the exploratory data analysis performed in Part I, this section identifies relationships between key variables such as **revenue, customer purchase frequency, unit price, product performance, and geographical location**.

The objective is to understand patterns, trends, and correlations that can support **data-driven decision-making** for business stakeholders through **interactive Tableau dashboards**.

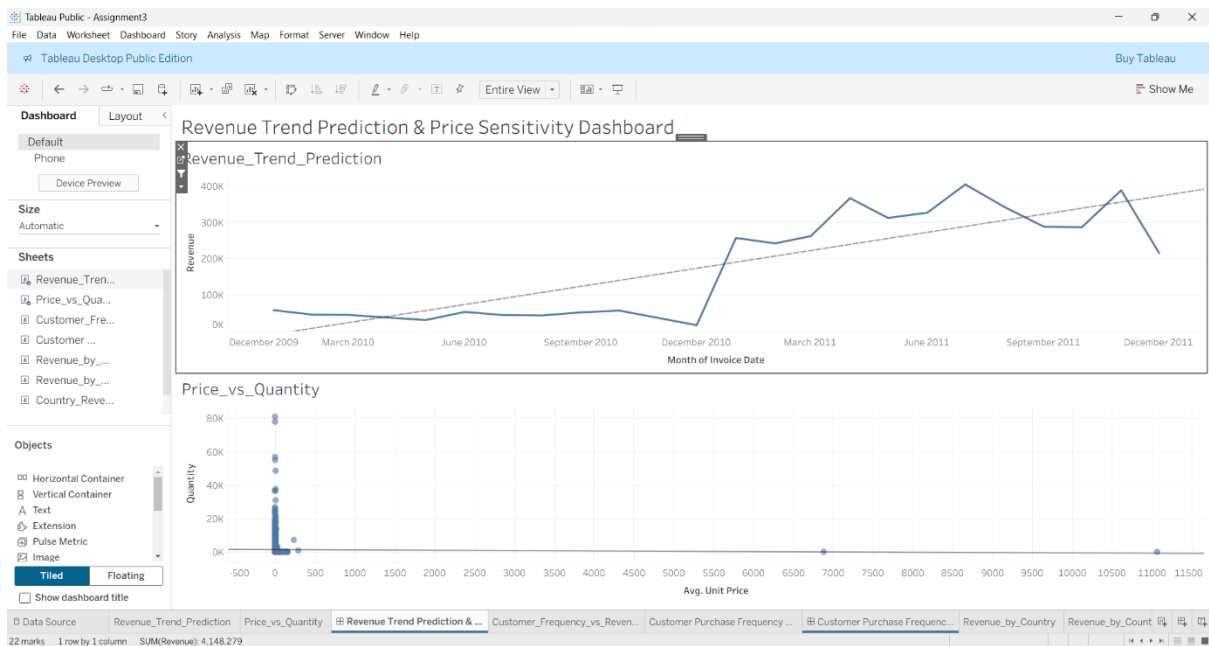
2. Identified Variable Relationships

The following key relationships were identified for predictive and prescriptive analysis:

1. **Time vs Revenue** – to identify revenue trends and forecast future performance
2. **Unit Price vs Quantity Sold** – to understand price sensitivity
3. **Customer Purchase Frequency vs Revenue** – to analyse customer value
4. **Product Performance vs Revenue Distribution** – to identify top-performing products
5. **Geographical Location vs Revenue** – to understand regional performance

Each relationship is visualised using dashboards designed for specific business audiences.

DASHBOARD 1: Revenue Trend Prediction & Price Sensitivity Dashboard



Target Audience

Senior Management and Sales Strategy Team

Purpose

To analyse historical revenue trends, identify growth patterns, and assess how pricing impacts sales quantity.

Information Presented

- Monthly revenue trend over time
- Trend line indicating future revenue direction
- Scatter plot showing relationship between average unit price and quantity sold

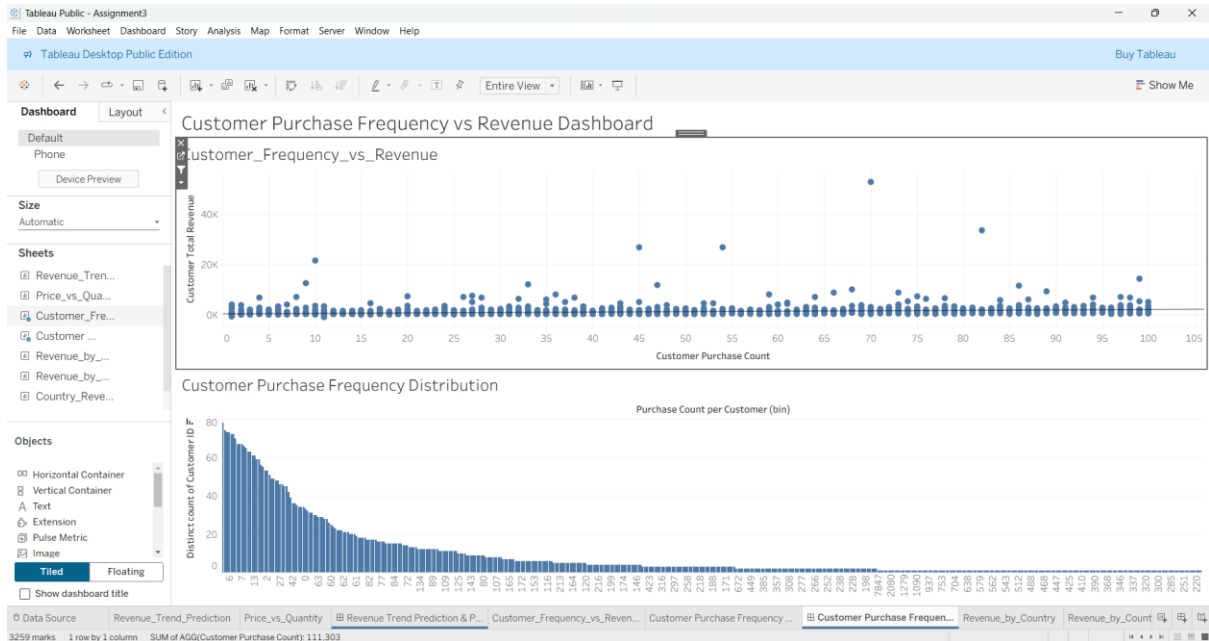
Key Insights

- Revenue shows an **overall upward trend**, indicating business growth
- Sharp increases suggest seasonal demand or successful campaigns
- Price vs quantity analysis indicates **price sensitivity**, where higher prices generally result in lower quantities sold

Actionable Insights & Decisions

- Plan inventory and marketing around high-growth periods
- Adjust pricing strategies to optimise revenue without negatively impacting volume

DASHBOARD 2: Customer Purchase Frequency vs Revenue Dashboard



Target Audience

Marketing Team and Customer Relationship Management (CRM) Team

Purpose

To understand how customer purchase frequency impacts total revenue contribution.

Information Presented

- Scatter plot of customer purchase count vs total customer revenue
- Distribution of customer purchase frequency using bins

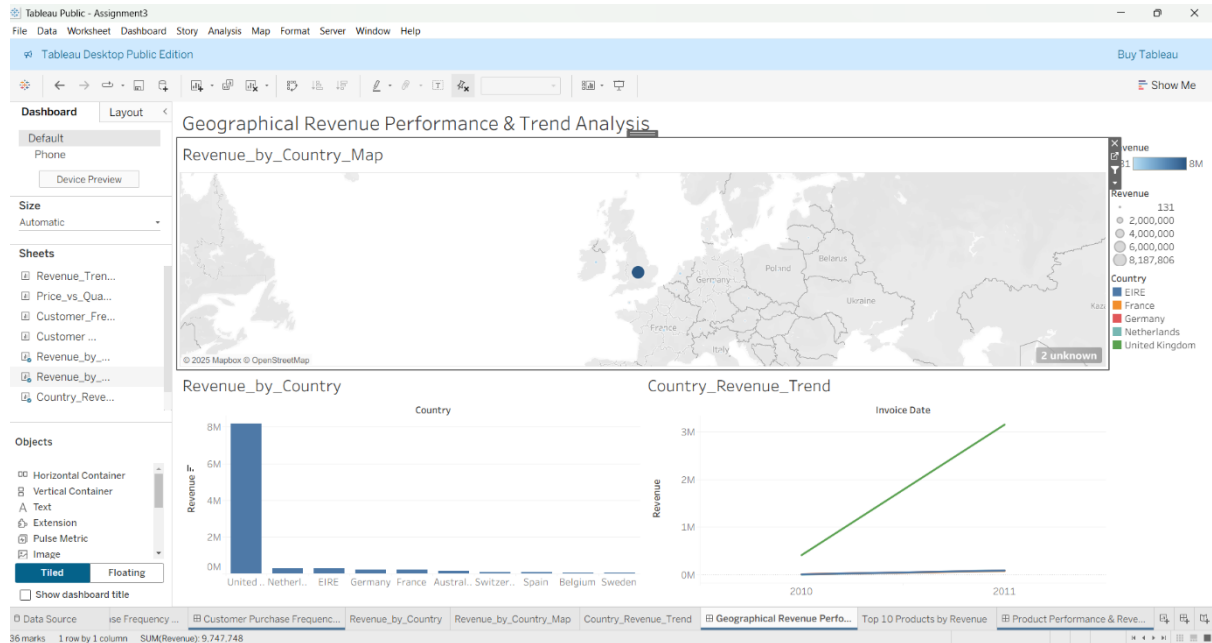
Key Insights

- A small group of high-frequency customers contributes disproportionately higher revenue
- Majority of customers are low-frequency buyers
- Clear positive relationship between purchase frequency and revenue

Actionable Insights & Decisions

- Identify and target high-value repeat customers
- Design loyalty programs to increase repeat purchases
- Segment customers for personalised marketing campaigns

DASHBOARD 3: Product Performance & Revenue Distribution Dashboard



Target Audience

Product Managers and Operations Team

Purpose

To evaluate product-level performance and identify top revenue-generating items.

Information Presented

- Top 10 products by revenue
- Revenue distribution across products

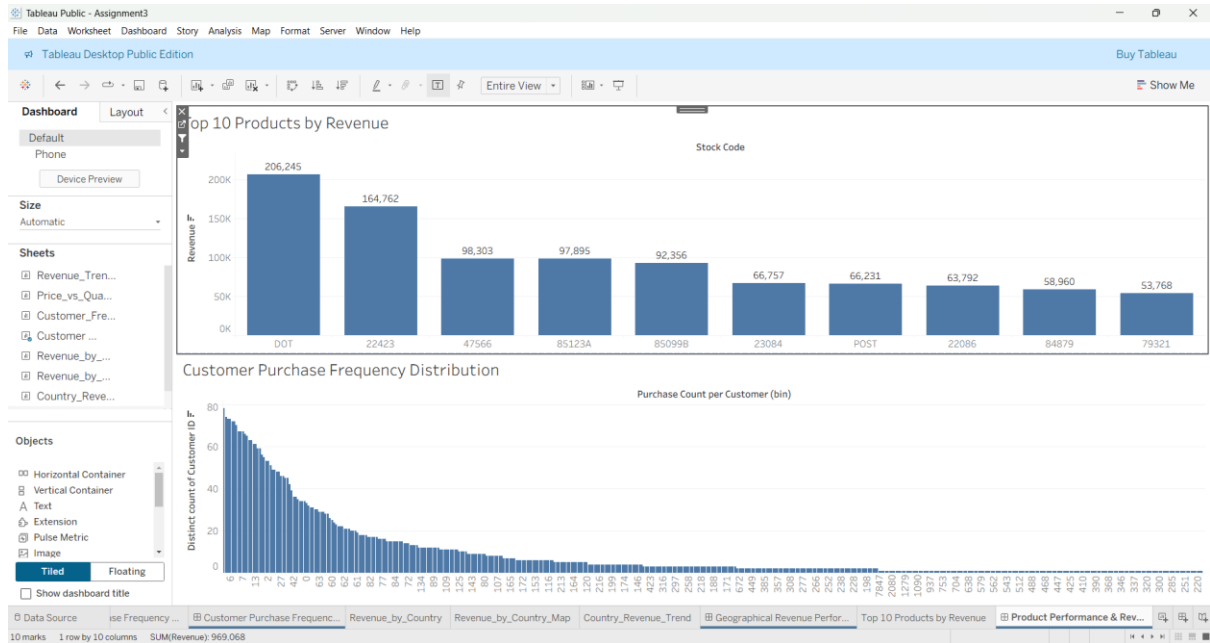
Key Insights

- A limited number of products generate the majority of revenue
- Long-tail products contribute minimal revenue individually
- Product performance follows a **Pareto (80/20) pattern**

Actionable Insights & Decisions

- Focus inventory and promotions on top-performing products
- Review underperforming products for discontinuation or repositioning
- Improve supply chain efficiency by prioritising high-revenue items

DASHBOARD 4: Geographical Revenue Performance Dashboard



Target Audience

Regional Sales Managers and Business Development Team

Purpose

To analyse revenue performance across different countries and track regional trends over time.

Information Presented

- Revenue by country displayed on a geographical map
- Country-wise revenue trend over time

Key Insights

- The United Kingdom dominates overall revenue contribution
- Other regions show comparatively lower but consistent growth
- Revenue trends vary significantly by country

Actionable Insights & Decisions

- Allocate more resources to high-performing regions
- Identify growth opportunities in underperforming markets
- Support regional expansion and localisation strategies

3. Statistical Modelling & Predictive Techniques

Trend lines and aggregation techniques were applied to model revenue growth and customer behaviour patterns. These techniques help estimate future performance and understand relationships between variables without overfitting complex models, ensuring clarity and interpretability for business users.

4. Decision-Making Benefits

The dashboards enable:

- Faster identification of revenue drivers
- Improved customer segmentation
- Better pricing and product strategies
- Data-backed regional expansion decisions

By visualising relationships and trends, the dashboards support **evidence-based decision-making** and reduce reliance on intuition.

5. Conclusion

This predictive analytics approach demonstrates how transactional sales data can be transformed into meaningful insights through visualisation. The four dashboards collectively provide strategic, tactical, and operational perspectives, enabling stakeholders to improve revenue planning, customer engagement, product performance, and regional growth.